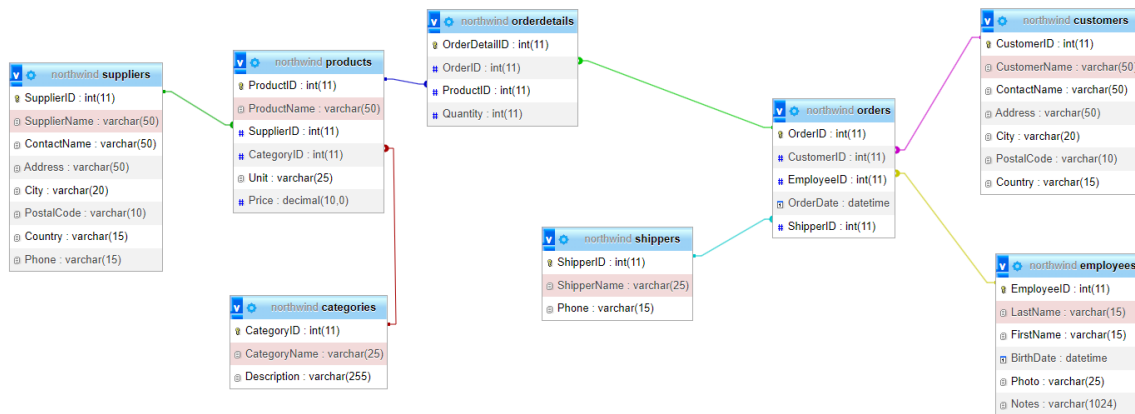
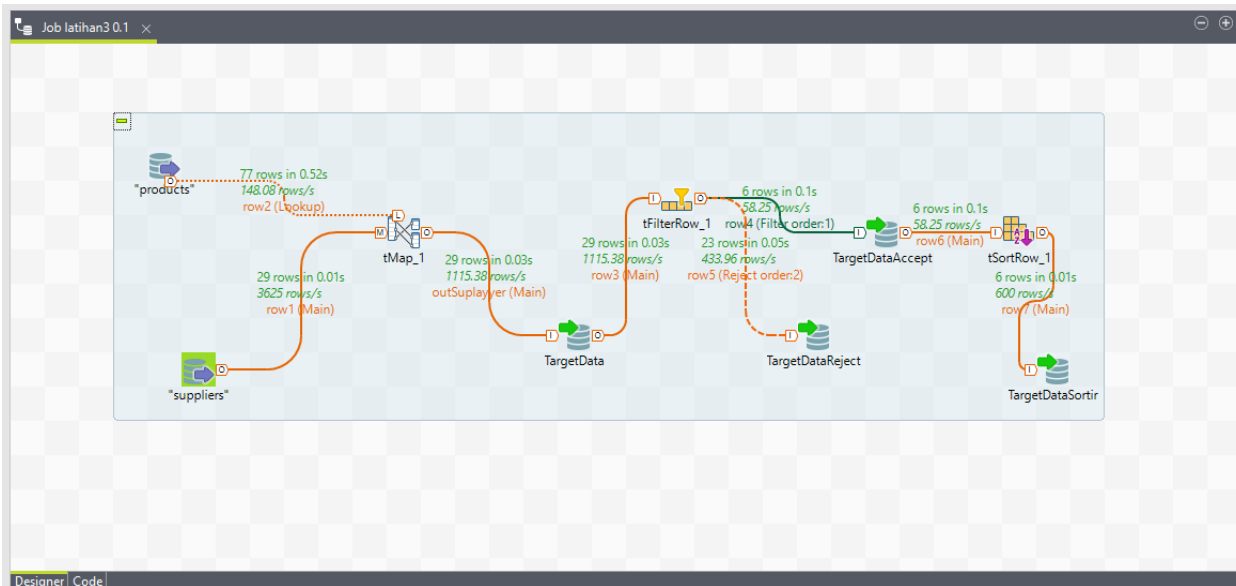


Praktikum Data Warehouse Petemuan 10

Relasi Tabel :



Skema :



===== Praktikum =====

1. Software Open Talend 7/8
2. Mysql
3. Aplikasi record layar (opsional)

===== Database =====

1. Database northwind (semua tabel)

===== Pratikum Membaca Data dari Metadata dan Filter Data =====

1. Buat koneksi MySQL dengan Sumber dan Target
Sumber (Northwind) :

Database Connection

New Database Connection on repository - Step 1/2

Define the properties

Name: SumberData

Purpose: Koneksi ke Database Northwind

Description: Digunakan sebagai sumberdata

Kemudian,

Database Connection

New Database Connection on repository - Step 2/2

Define the connection parameters

DB Type: MySQL

Db Version: MySQL 5

String of Connection: jdbc:mysql://localhost:3306/northwind?noDatetimeStringSync=true

Login: root

Password:

Server: localhost

Port: 3306

DataBase: northwind

Additional parameters: noDatetimeStringSync=true

Test connection

2. Kemudian buat Tabel Skema Data Pada Sumber Data.

Schema

Filter for the Table.

Select Filter Conditions

☒ Use the Name Filter ☐ Use the Sql Filter

Select Types

☒ TABLE ☐ VIEW ☐ SYNONYM

Set the Name Filter:

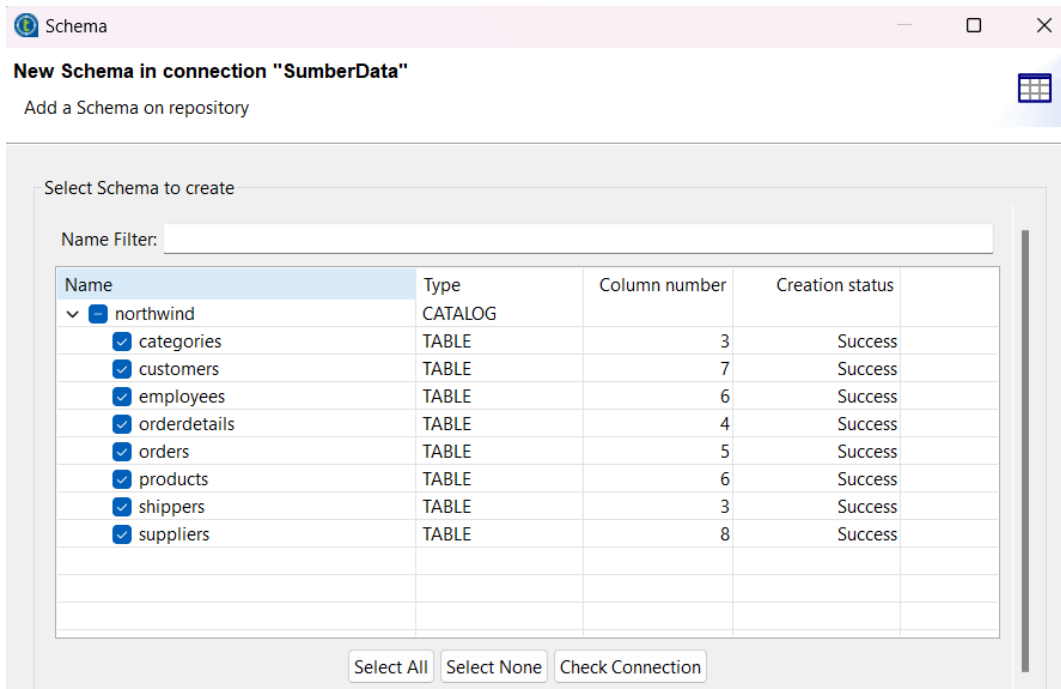
%

New...

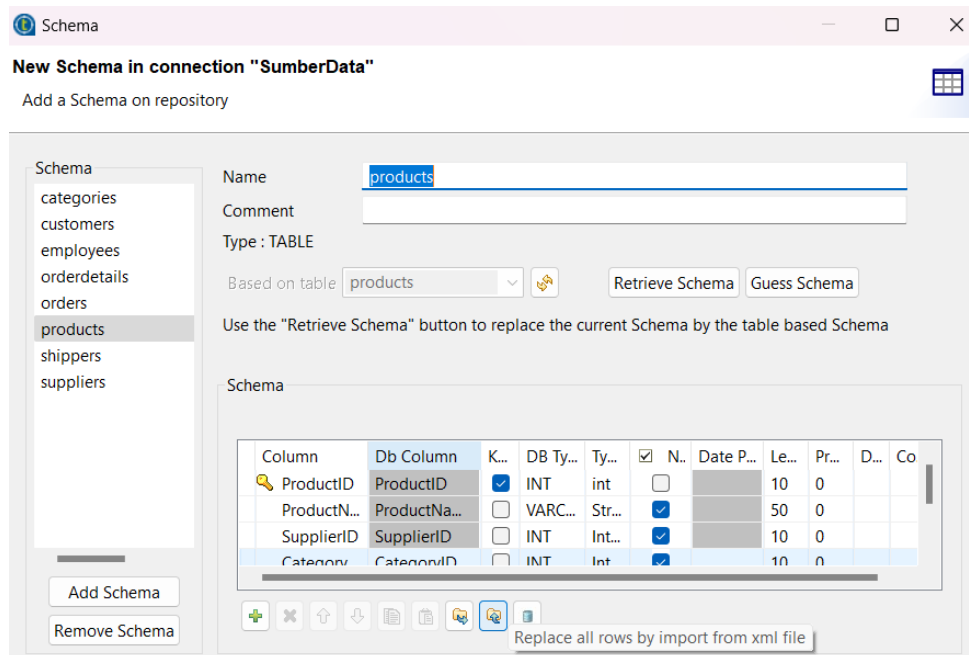
Edit...

Remove...

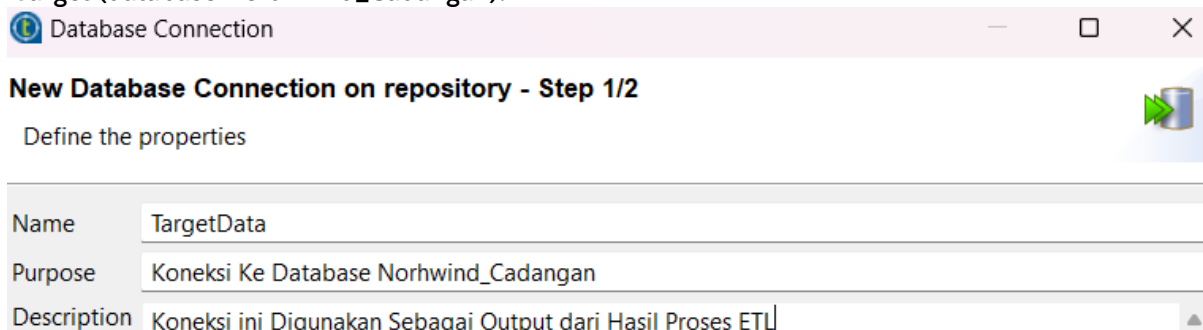
Kemudian ambil semua table pada db northwind,



Kemudian,



3. Buat dan lakukan koneksi ke MySQL Target
Target (database Northwind_Cadangan):



Kemudian,

Database Connection

New Database Connection on repository - Step 2/2

Define the connection parameters

DB Type: MySQL

Db Version: MySQL 5

String of Connection: jdbc:mysql://localhost:3306/northwind_cadangan?noDatetimeStringSync=true

Login: root

Password:

Server: localhost

Port: 3306

DataBase: northwind_cadangan

Additional parameters: noDatetimeStringSync=true

Test connection

4. Buatlah Job Design Baru untuk melihat informasi dari Metadata (*nama job design disesuaikan*)

New Job

New Job

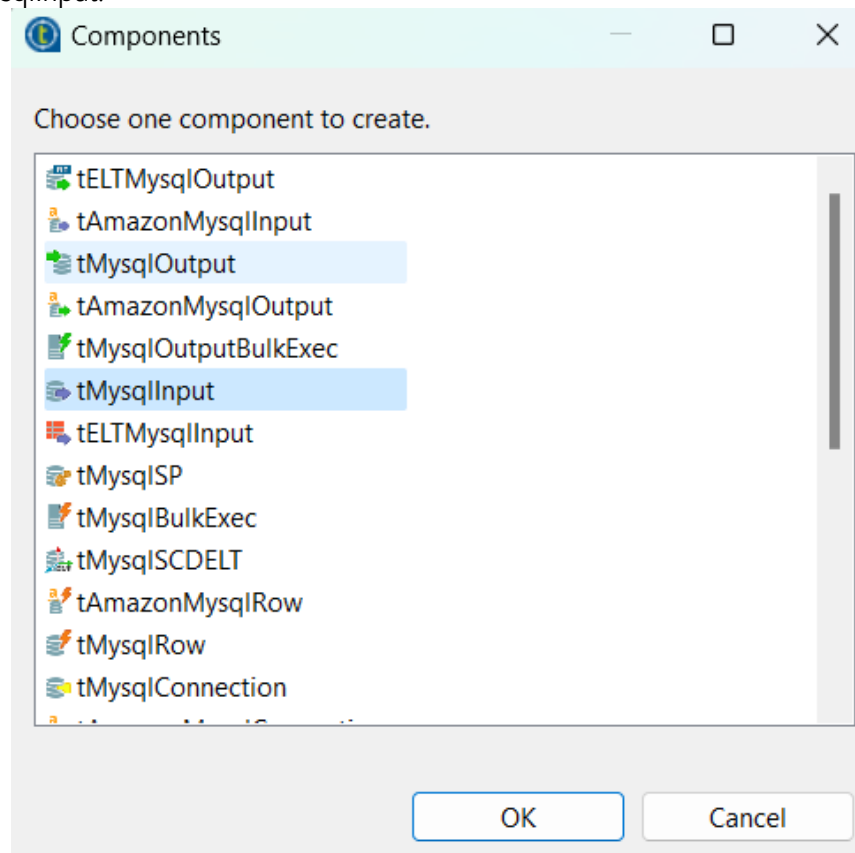
Add a job in the repository

Name: Latihan1

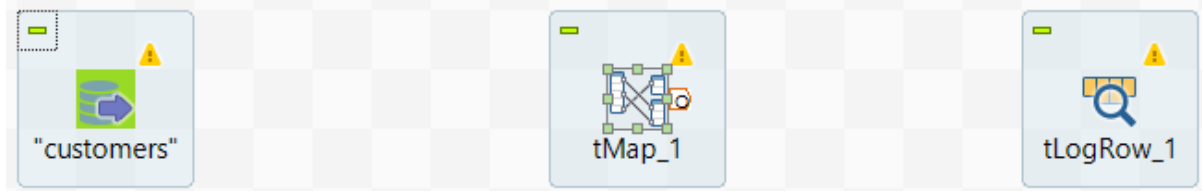
Purpose: Cek perintah dasar melihat dari data metadata

Description: Menggunakan tInputMysql, tMap, tLogRow

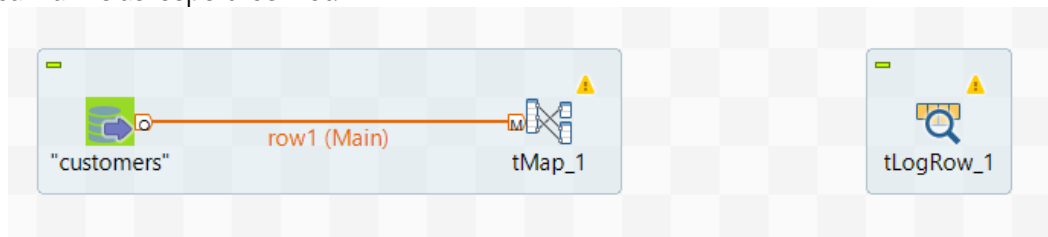
5. Masukan data table customers dengan menggunakan cara **drag and drop** ke bagian job design. Lalu gunakan tMySQLInput.



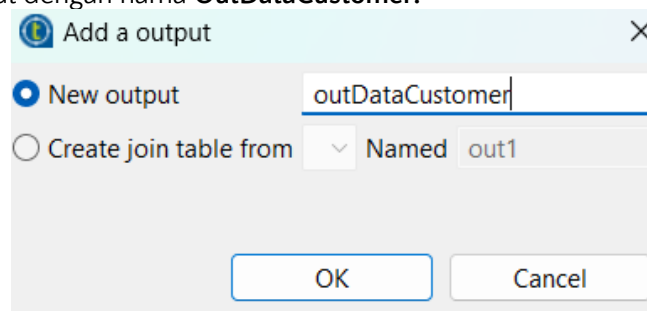
6. Masukan tools berikut: **tMap** dan **tLogRow**



7. Tambahkan relasi seperti berikut.



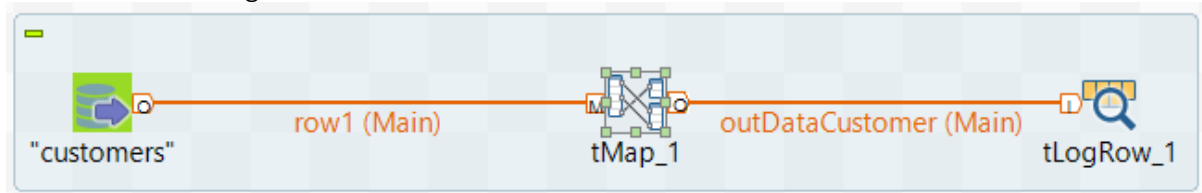
8. Kemudian Buat Output dengan nama **OutDataCustomer**.



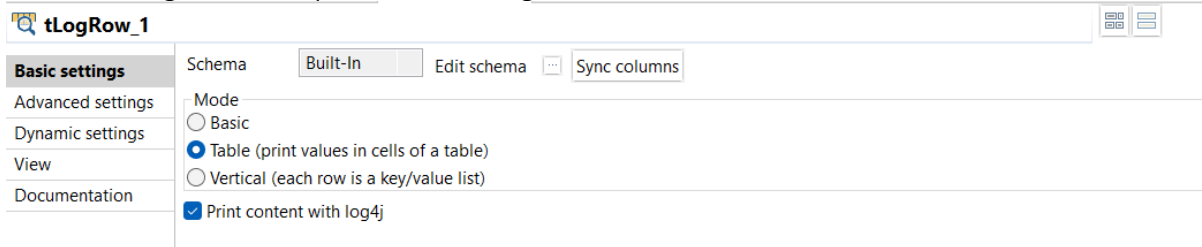
9. Kemudian pada tMap buat skema berikut dan apply.



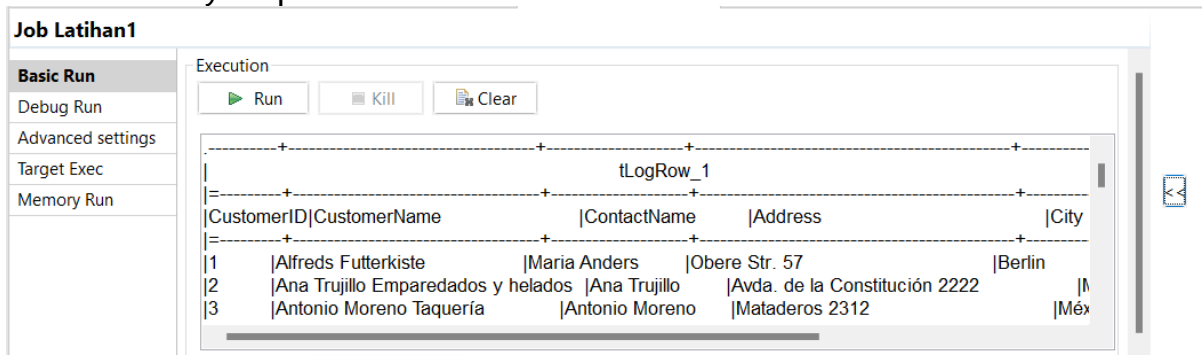
10. Buat relasi ke tLogRow.



11. Lakukan Pengaturan component untuk tLogRow kemudian Run.

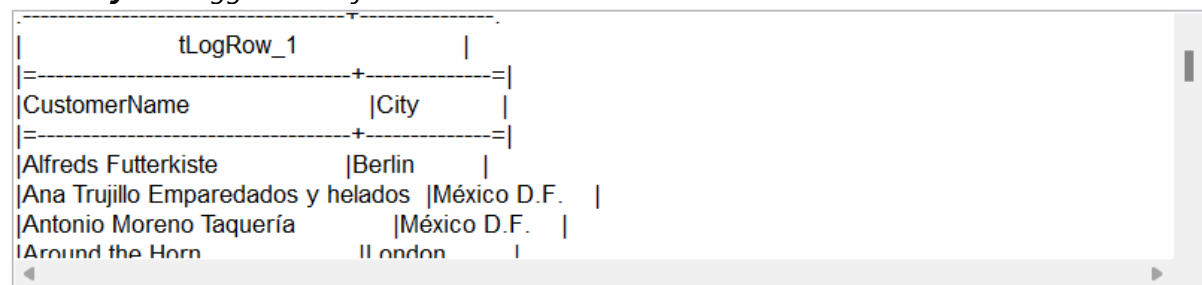


Pastikan hasil dilayar seperti berikut :

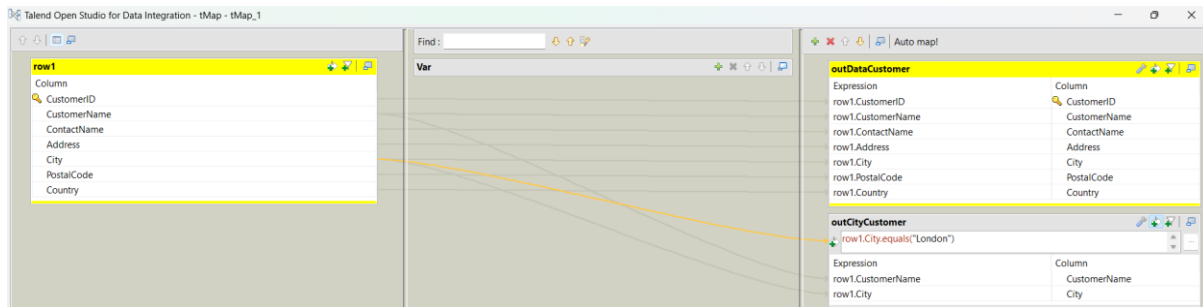


===== Tugas 1 =====

1. Buat Output dengan nama **OutCityCustomer** dengan hanya menggunakan data **CustomerName** dan **City**. Sehingga menunjukkan hasil berikut:



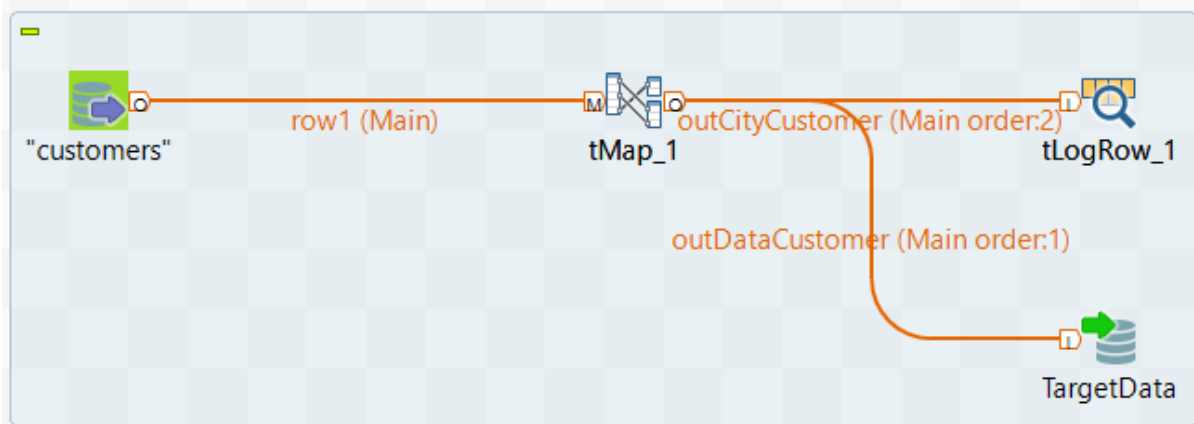
12. Selanjutnya lakukan Filter data dengan nama kota London pada Output **OutCityCustomer** dengan menambahkan kondisi **row1.City.equals(London)** seperti berikut:



Dengan hasil Ketika di Run:

tLogRow_1	
CustomerName	City
Around the Horn	London
B's Beverages	London
Consolidated Holdings	London
Eastern Connection	London

13. Masukan target data dan berikan relasi menggunakan tMySQLOutput untuk Output **OutDataCustomer**.



Kemudian lakukan pengaturan component **Target Data** seperti berikut.

TargetData(tDBOutput_1)(MySQL)

Basic settings	Database: MySQL	Apply
Advanced settings	Property Type: Repository	DB (MYSQL):TargetData
Dynamic settings	DB Version: Mysql 5	
View	☐ Use an existing connection	
Documentation	Host: "localhost"	Port: "3306"
	Database: "northwind_cadangan"	
	Username: "root"	Password: "*****"
	Table: "Data Customers"	
	Action on table: Drop and create table	Action on data: Insert
	Schema: Built-In	Edit schema ... Sync columns

Kemudian Run, pastikan pada database **northwind_cadangan** terdapat table **Data Customers** dengan tampilan berikut (sesuaikan dengan nama database dan table masing-masing).

		CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
Edit Copy Delete		1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
Edit Copy Delete		2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	5021	Mexico
Edit Copy Delete		3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	5023	Mexico
Edit Copy Delete		4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
Edit Copy Delete		5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
Edit Copy Delete		6	Blauer See Delikatessen	Hanna Moos	Forsterstr. 57	Mannheim	68306	Germany
Edit Copy Delete		7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	67000	France
Edit Copy Delete		8	Bólido Comidas preparadas	Martin Sommer	C/ Araquil, 67	Madrid	28023	Spain
Edit Copy Delete		9	Bon app"	Laurence Lebihans	12, rue des Bouchers	Marseille	13008	France
Edit Copy Delete		10	Bottom-Dollar Marketse	Elizabeth Lincoln	23 Tsawassen Blvd.	Tsawassen	T2F 8M4	Canada
Edit Copy Delete		11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
Edit Copy Delete		12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
Edit Copy Delete		13	Centro comercial Moctezuma	Francisco Chang	Sierras de Granada 9993	México D.F.	5022	Mexico
Edit Copy Delete		14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
Edit Copy Delete		15	Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	São Paulo	05432-043	Brazil
Edit Copy Delete		16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
Edit Copy Delete		17	Drachenblut Delikatessend	Sven Ottlieb	Walservweg 21	Aachen	52066	Germany
Edit Copy Delete		18	Du monde entier	Janine Labrune	67, rue des Cinquante Otages	Nantes	44000	France
Edit Copy Delete		19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
Edit Copy Delete		20	Ernst Handel	Roland Mendel	Kirchgasse 6	Graz	8010	Austria
Edit Copy Delete		21	Familia Arquibaldo	Aria Cruz	Rua Orós, 92	São Paulo	05442-030	Brazil
Edit Copy Delete		22	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel	C/ Moralzarzal, 86	Madrid	28034	Spain
Edit Copy Delete		23	Folies gourmandes	Martine Rancé	184, chaussée de Tournai	Lille	59000	France
Edit Copy Delete		24	Folk och fa HB	Maria Larsson	Åkergatan 24	Bräcke	S-844 67	Sweden
Edit Copy Delete		25	Frankenversand	Peter Franken	Berliner Platz 43	München	80805	Germany

===== Tugas 2 =====

1. Buat Output baru (nama output disesuaikan) untuk filter data Customer dengan Kota **"Madrid"** dan dihubungkan dengan TargetData **Tabel DataCustomerMadrid** dengan pengaturan berikut:

TargetData(tDBOutput_1)(MySQL)

Basic settings

Advanced settings

Dynamic settings

View

Documentation

Database

MySQL

Apply

Property Type

Repository

DB (MYSQL):TargetData

DB Version

MySQL 5

Use an existing connection

☐

Host

localhost

Port

3306

Database

northwind_cadangan

The variable attached to this parameter is: __HOST__

Username

root

Password

Table

DataCustomerMadrid

Action on table

Drop table if exists and create

Action on data

Insert

Schema

Built-In

Edit schema

Sync columns

Dengan hasil Ketika di Run:

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds.)

SELECT * FROM `datacustomermadrid`

Profiling

Edit inline

Edit

Explain SQL

Create PHP code

Refresh

Show all

Number of rows:

25

Filter rows:

Search this table

Extra options

CustomerName

City

Bólido Comidas preparadas

Madrid

FISSA Fabrica Inter. Salchichas S.A.

Madrid

Romero y tomillo

Madrid

Show all

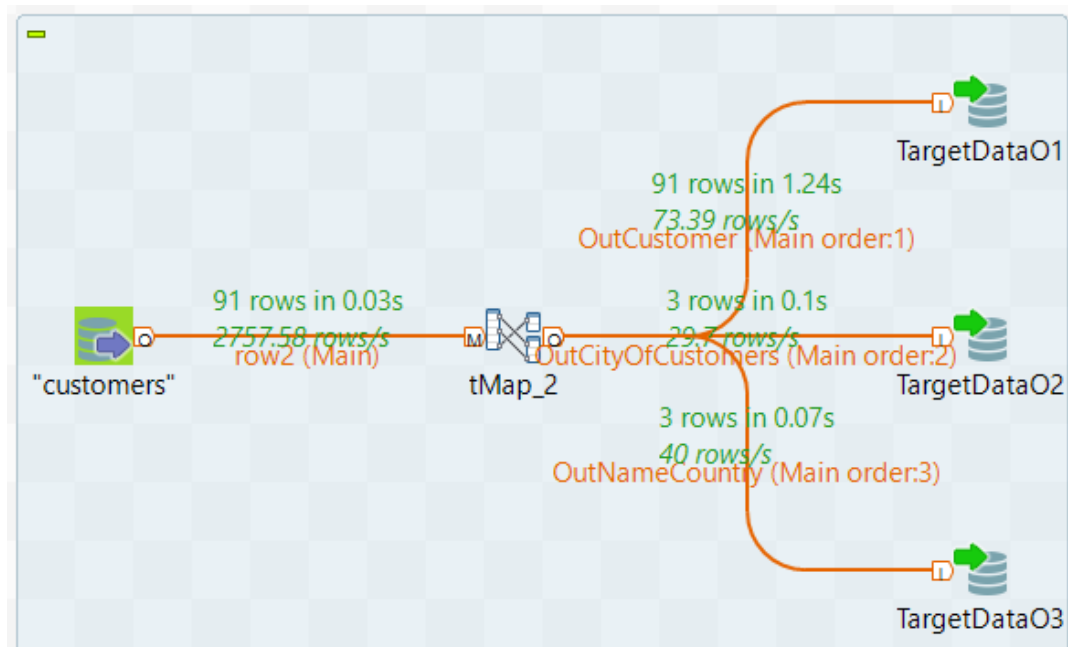
Number of rows:

25

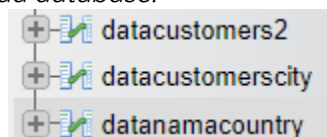
Filter rows:

Search this table

2. Buat Job Design baru dengan filter data dengan kota Madrid dan Canada, dengan contoh skema:



Dengan hasil Ketika di Run, Tabel pada database:



Data pada tabel datacustomer2 :

	CustomerID	CustomerName	ContactName	Address	City	PostalCode	Country
☐ Edit Copy Delete	1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209	Germany
☐ Edit Copy Delete	2	Ana Trujillo Emparedados y helados	Ana Trujillo	Avda. de la Constitución 2222	México D.F.	5021	Mexico
☐ Edit Copy Delete	3	Antonio Moreno Taquería	Antonio Moreno	Mataderos 2312	México D.F.	5023	Mexico
☐ Edit Copy Delete	4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	WA1 1DP	UK
☐ Edit Copy Delete	5	Berglunds snabbköp	Christina Berglund	Berguvsvägen 8	Luleå	S-958 22	Sweden
☐ Edit Copy Delete	6	Blauer See Delikatessen	Hanna Moos	Forsterstr. 57	Mannheim	68306	Germany
☐ Edit Copy Delete	7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg	67000	France
☐ Edit Copy Delete	8	Bólido Comidas preparadas	Martin Sommer	C/ Araquil, 67	Madrid	28023	Spain
☐ Edit Copy Delete	9	Bon app'	Laurence Leblhans	12, rue des Bouchers	Marseille	13008	France
☐ Edit Copy Delete	10	Bottom-Dollar Marketse	Elizabeth Lincoln	23 Tsawassen Blvd.	Tsawassen	T2F 8M4	Canada
☐ Edit Copy Delete	11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC2 5NT	UK
☐ Edit Copy Delete	12	Cactus Comidas para llevar	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
☐ Edit Copy Delete	13	Centro comercial Moctezuma	Francisco Chang	Sierras de Granada 9993	México D.F.	5022	Mexico
☐ Edit Copy Delete	14	Chop-suey Chinese	Yang Wang	Hauptstr. 29	Bern	3012	Switzerland
☐ Edit Copy Delete	15	Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	São Paulo	05432-043	Brazil
☐ Edit Copy Delete	16	Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	London	WX1 6LT	UK
☐ Edit Copy Delete	17	Drachenblut Delikatessend	Sven Ottlieb	Walserweg 21	Aachen	52066	Germany
☐ Edit Copy Delete	18	Du monde entier	Janine Labrune	67, rue des Cinquante Otages	Nantes	44000	France
☐ Edit Copy Delete	19	Eastern Connection	Ann Devon	35 King George	London	WX3 6FW	UK
☐ Edit Copy Delete	20	Ernst Handel	Roland Mendel	Kirchgasse 6	Graz	8010	Austria
☐ Edit Copy Delete	21	Familia Arquibaldo	Aria Cruz	Rua Orós, 92	São Paulo	05442-030	Brazil
☐ Edit Copy Delete	22	FISSA Fabrica Inter. Salchichas S.A.	Diego Roel	C/ Moralzarzal, 86	Madrid	28034	Spain
☐ Edit Copy Delete	23	Folies gourmandes	Martine Rancé	184, chaussée de Tournai	Lille	59000	France
☐ Edit Copy Delete	24	Folk och fa HB	Maria Larsson	Åkergatan 24	Bräcke	S-844 67	Sweden
☐ Edit Copy Delete	25	Frankenversand	Peter Franken	Berliner Platz 43	München	80805	Germany

Data pada tabel datacustomerscity :

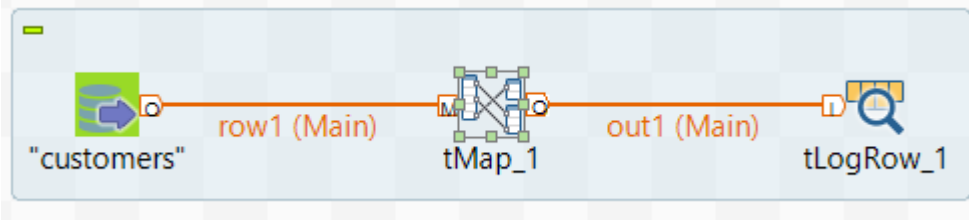
CustomerName	City
Bólido Comidas preparadas	Madrid
FISSA Fabrica Inter. Salchichas S.A.	Madrid
Romero y tomillo	Madrid

Data pada tabel datanamacountry :

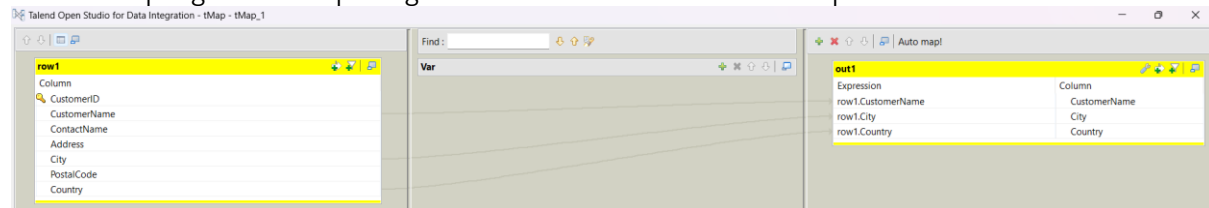
CustomerName	Country
Bottom-Dollar Marketse	Canada
Laughing Bacchus Wine Cellars	Canada
Mère Paillarde	Canada

===== Pratikum Membuat Variabel =====

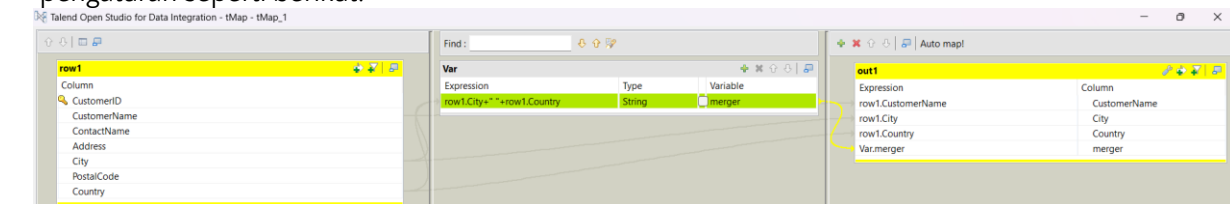
1. Buat Job Design Baru dan masukkan table customer dengan drag and drop, tMap, dan tLogRow.



2. Lakukan pengaturan tMap dengan double click dan buat skema seperti berikut:



3. Buat variable hasil penggabungan data dengan klik tombol + pada **Var** (kotak ditengah) dan lakukan pengaturan seperti berikut:



Tambahkan expression (kondisi) untuk menggabungkan 2 data dengan double klik seperti berikut:

Expression: `row1.City+" "+row1.Country`

Test: `row1.Custo... 0`

Categories: *All, *User Defined, DataOperation, Mathematical, Numeric, Relational

Functions: (empty)

Help: Please select a category and function.

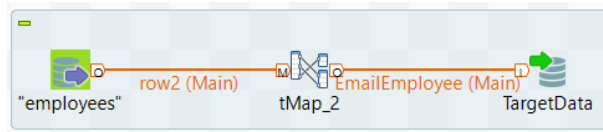
Buttons: Ok, Cancel

4. Lakukan Run dan pastikan menampilkan hasil seperti berikut:

tLogRow_1			
CustomerName	City	Country	merger
Alfreds Futterkiste	Berlin	Germany	Berlin Germany
Ana Trujillo Emparedados y helados	México D.F.	Mexico	México D.F. Mexico
Antonio Moreno Taquería	México D.F.	Mexico	México D.F. Mexico

===== Tugas 3 =====

1. Buat Job Design baru dan masukkan tMysqlInput (tabel employees) dan tMap, kemudian buat Variabel **Email** dengan menggabungkan **row.FirstName** dan **row.LastName** menjadi satu email baru dengan hasil:



	EmployeeID	LastName	FirstName	BirthDate	Photo	Notes	Email
☐ Edit ☐ Copy ☐ Delete	1	Davolio	Nancy	1968-12-08 00:00:00	EmplD1.pic	Education includes a BA in psychology from Colorado State University in 1992. Completed the management PMP certified course in 1993. Completed the sales management within sales training in 1994 and the marketing department training in 1995. Completed the Strategic Operations Management training in 1996.	Nancy.Davolio@himatikom.id
☐ Edit ☐ Copy ☐ Delete	2	Fuller	Andrew	1952-02-19 00:00:00	EmplD2.pic	Andrew received his BTS commercial and a Ph.D. in Marketing from the University of South Alabama in 1975. Received the Marketing Executive Certificate from the University of Southern California in 1982.	Andrew.Fuller@himatikom.id
☐ Edit ☐ Copy ☐ Delete	3	Leverling	Janet	1963-08-30 00:00:00	EmplD3.pic	Janet has a BS degree in chemistry from Boston College in 1984. She then completed her M.S. in chemistry from Boston College in 1985.	Janet.Leverling@himatikom.id
☐ Edit ☐ Copy ☐ Delete	4	Peacock	Margaret	1958-09-19 00:00:00	EmplD4.pic	Margaret holds a BA in English literature from Colorado State University in 1982. She then completed her M.S. in English literature from Colorado State University in 1984.	Margaret.Peacock@himatikom.id
☐ Edit ☐ Copy ☐ Delete	5	Buchanan	Steven	1955-03-04 00:00:00	EmplD5.pic	Steven Buchanan graduated from St. Andrews University in Scotland in 1982. Completed the Strategic Operations Management training in 1996.	Steven.Buchanan@himatikom.id
☐ Edit ☐ Copy ☐ Delete	6	Suyama	Michael	1963-07-02 00:00:00	EmplD6.pic	Michael is a graduate of Sussex University (MA, Essex, UK) in 1982. He completed his second degree in Chemistry at Sussex University in 1985.	Michael.Suyama@himatikom.id
☐ Edit ☐ Copy ☐ Delete	7	King	Robert	1960-05-29 00:00:00	EmplD7.pic	Robert King served in the Peace Corps and traveled the world. Completed the Strategic Operations Management training in 1996.	Robert.King@himatikom.id
☐ Edit ☐ Copy ☐ Delete	8	Callahan	Laura	1958-01-09 00:00:00	EmplD8.pic	Laura received a BA in psychology from the University of Iowa in 1982. She then completed her M.S. in psychology from the University of Iowa in 1984.	Laura.Callahan@himatikom.id
☐ Edit ☐ Copy ☐ Delete	9	Dodsworth	Anne	1969-07-02 00:00:00	EmplD9.pic	Anne has a BA degree in English from St. Lawrence University in 1982. She then completed her M.S. in English from St. Lawrence University in 1984.	Anne.Dodsworth@himatikom.id
☐ Edit ☐ Copy ☐ Delete	10	West	Adam	1928-09-19 00:00:00	EmplD10.pic	An old chum.	Adam.West@himatikom.id

===== Pratikum Sortir Data =====

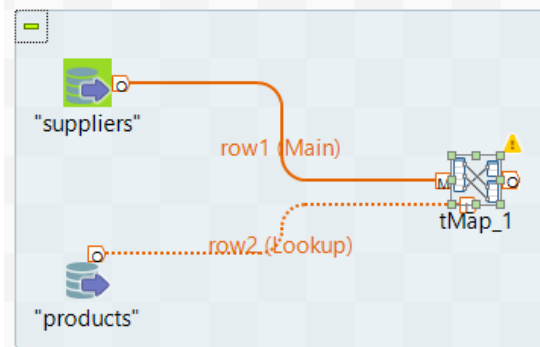
1. Buat Job Design baru untuk praktikum Sortir Data

New Job
Add a job in the repository

Name: Latihan3
Purpose: Membuat Filter dan Shorting
Description: -
Author: user@talend.com
Locker:
Version: 0.1
Status:
Path:
Select

Finish Cancel

2. Buat ske berikut dengan gunakan tMySQLInput dari metadata supplier dan product.



3. Lakukan pengaturan tMap, seperti berikut:

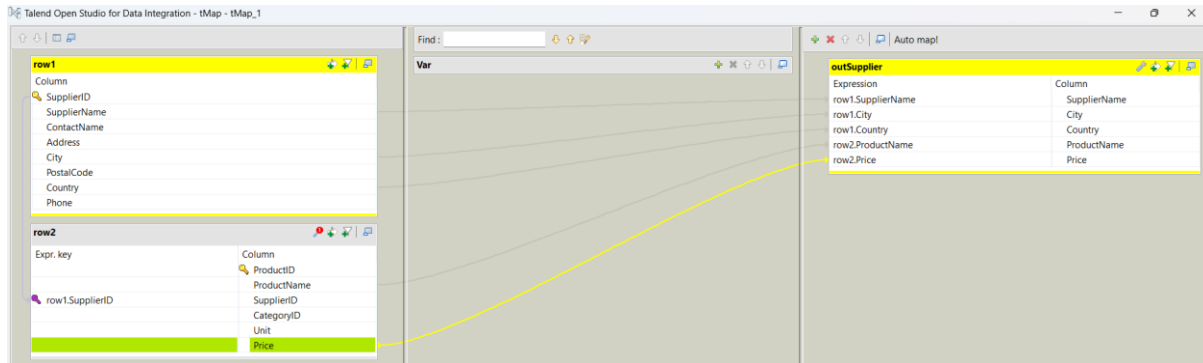
row1
Column
SupplierID
SupplierName
ContactName
Address
City
PostalCode
Country
Phone

row2
Property Value
Lookup Model Load once
Match Model All rows
Join Model Left Outer Join
Store temp data false
Expr. key
row1.SupplierID
ProductID
ProductName
SupplierID
CategoryID
Unit
Price

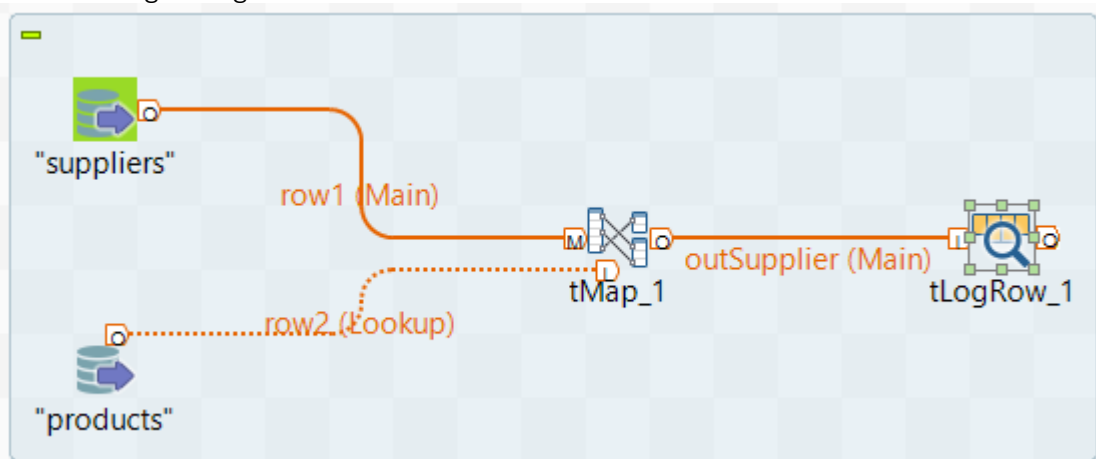
Var
ProductID
ProductName
SupplierID
CategoryID
Unit
Price

Auto map!

Tambahkan tLogRow dan buat output baru **OutSupplier**. Tambahkan skema berikut pada tmap (Masukkan beberapa data dari row1 dan row2 ke outSupplier).



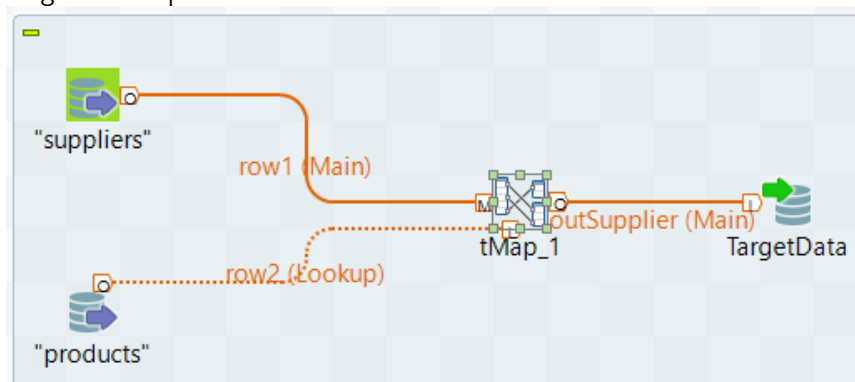
4. Buat relasi dengan tLogRow dan lakukan Run.



Pastikan hasil running seperti berikut:

tLogRow_1				
SupplierName	City	Country	ProductName	Price
Exotic Liquid	Londona	UK	Aniseed Syrup	10
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40

5. Tambahkan TargetData seperti berikut.



Lakukan pengaturan TargetData seperti berikut:

TargetData(tDBOutput_1)(MySQL)

Basic settings

Database: MySQL Apply

Property Type: Repository DB (MYSQL):TargetData

Dynamic settings

DB Version: Mysql 5

☐ Use an existing connection

Host: "localhost" Port: "3306"

Database: "northwind_cadangan"

Username: "root" Password: "*****"

Table: "DataSupplier1"

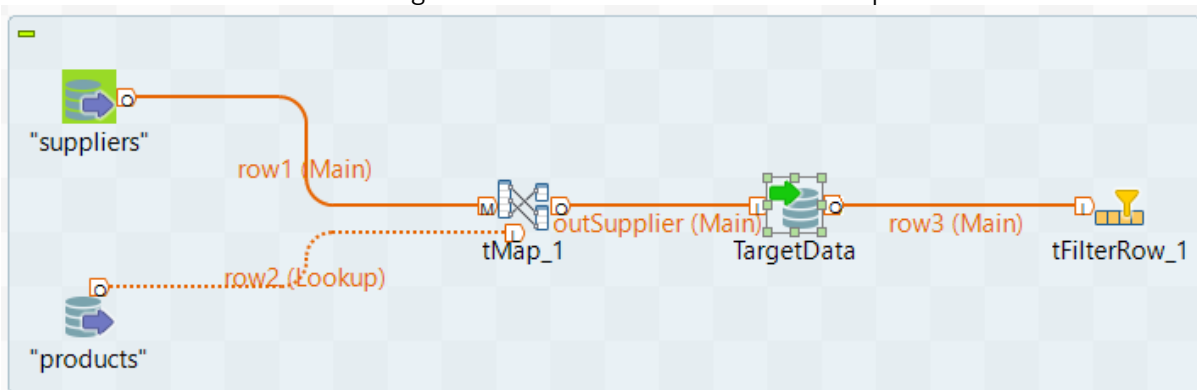
Action on table: Drop table if exists and create Action on data: Insert

Schema: Built-In Edit schema Sync columns

Lakukan Run dan pastikan hasil seperti berikut:

SupplierName	City	Country	ProductName	Price
Exotic Liquid	Londona	UK	Aniseed Syrup	10
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40
Tokyo Traders	Tokyo	Japan	Longlife Tofu	10
Cooperativa de Quesos 'Las Cabras'	Oviedo	Spain	Queso Manchego La Pastora	38
Mayumi's	Osaka	Japan	Genen Shouyu	16
Pavlova, Ltd.	Melbourne	Australia	Outback Lager	15
Specialty Biscuits, Ltd.	Manchester	UK	Scottish Longbreads	13
PB Knäckebröd AB	Göteborg	Sweden	Tunnbröd	9
Refrescos Americanas LTDA	São Paulo	Brazil	Guaraná Fantástica	5
Heli Süßwaren GmbH & Co. KG	Berlin	Germany	Schoggi Schokolade	44
Plutzer Lebensmittelgroßmärkte AG	Frankfurt	Germany	Original Frankfurter grüne Soße	13
Nord-Ost-Fisch Handelsgesellschaft mbH	Cuxhaven	Germany	Nord-Ost Matjeshering	26
Formaggi Fortini s.r.l.	Ravenna	Italy	Mozzarella di Giovanni	35
Norske Meierier	Sandvika	Norway	Fløtemysost	22
Bigfoot Breweries	Bend	USA	Laughing Lumberjack Lager	14
Svensk Sjöföda AB	Stockholm	Sweden	Röd Kaviar	15
Aux joyeux ecclésiastiques	Paris	France	Chartreuse verte	18
New England Seafood Cannery	Boston	USA	Jack's New England Clam Chowder	10
Leka Trading	Singapore	Singapore	Gula Malacca	19
Lyngbysild	Lyngby	Denmark	Spegesild	12
Zaanse Snoepfabriek	Zaandam	Netherlands	Chocolade	13
Karkki Oy	Lappeenranta	Finland	Lakkalikööri	18
G'day, Mate	Sydney	Australia	Perth Pasties	33
Ma Maison	Montréal	Canada	Pâté chinois	24

6. Kemudian lakukan filter data dengan menambahkan tools tFilterRow seperti berikut.



Lakukan pengaturan component tFilterRow untuk filter Negara USA seperti berikut.

tFilterRow_1

Schema: Built-In Edit schema Sync columns

Logical operator used to combine conditions: And

InputColumn	Function	Operator	Value
Country	Empty	Equals	"USA"

Use advanced mode

7. Tambahkan tLogRow dan Lakukan Run. Pastikan menunjukkan hasil berikut:

Execution

Run Kill Clear

tLogRow_1				
SupplierName	City	Country	ProductName	Price
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40
Bigfoot Breweries	Bend	USA	Laughing Lumberjack Lager	14
New England Seafood Cannery	Boston	USA	Jack's New England Clam Chowder	10

===== Tugas 4 =====

Tambahkan Filter Nagara "UK" pada tFilterRow dan Run. Pastikan hasilnya seperti berikut:

Job(Latihan3 0.1) Contexts(Latihan3) Component Run (Job Latihan3)

Job Latihan3

Basic Run Debug Run Advanced settings Target Exec Memory Run

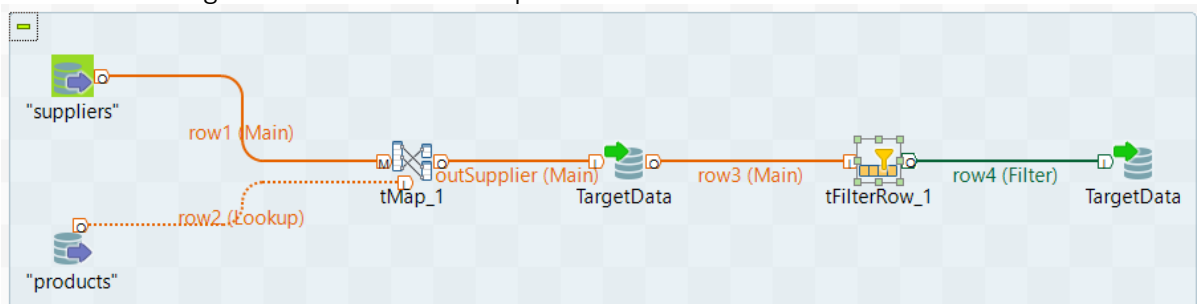
Execution

Run Kill Clear

tLogRow_1				
SupplierName	City	Country	ProductName	Price
Exotic Liquid	Londona	UK	Aniseed Syrup	10
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40

Line limit: 100 Wrap

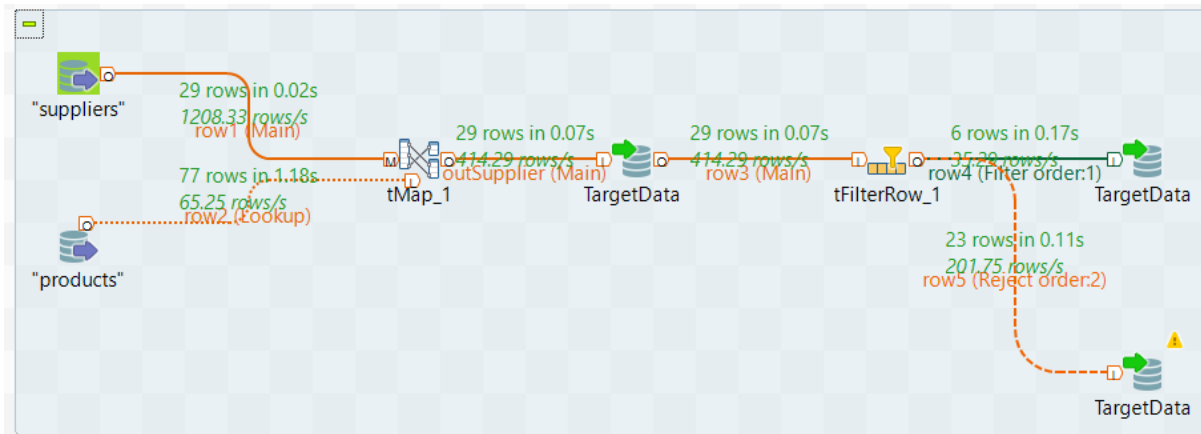
8. Tambahkan TargetData dan buat relasi seperti berikut:



Lakukan Run dan Pastikan hasilnya seperti berikut:

SupplierName	City	Country	ProductName	Price
Exotic Liquid	Londona	UK	Aniseed Syrup	10
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40
Specialty Biscuits, Ltd.	Manchester	UK	Scottish Longbreads	13
Bigfoot Breweries	Bend	USA	Laughing Lumberjack Lager	14
New England Seafood Cannery	Boston	USA	Jack's New England Clam Chowder	10

9. Tambahkan Target data dengan menambah relasi reject seperti berikut:

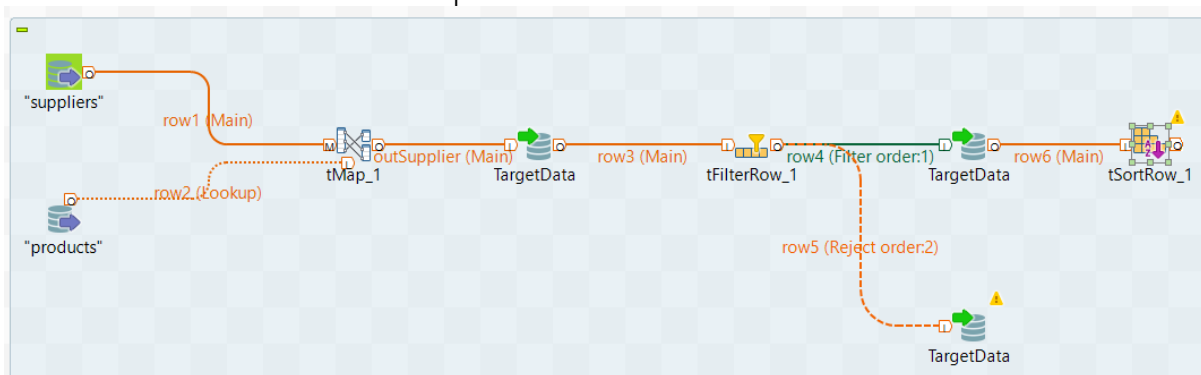


Lakukan Run dan Pastikan hasilnya seperti berikut:

Extra options

SupplierName	City	Country	ProductName	Price
Tokyo Traders	Tokyo	Japan	Longlife Tofu	10
Cooperativa de Quesos 'Las Cabras'	Oviedo	Spain	Queso Manchego La Pastora	38
Mayumi's	Osaka	Japan	Genen Shouyu	16
Pavlova, Ltd.	Melbourne	Australia	Outback Lager	15
PB Knäckebröd AB	Göteborg	Sweden	Tunnbröd	9
Refrescos Americanas LTDA	São Paulo	Brazil	Guaraná Fantástica	5
Hell Süßwaren GmbH & Co. KG	Berlin	Germany	Schoggi Schokolade	44
Plutzer Lebensmittelgroßmärkte AG	Frankfurt	Germany	Original Frankfurter grüne Soße	13
Nord-Ost-Fisch Handelsgesellschaft mbH	Cuxhaven	Germany	Nord-Ost Matjeshering	26
Formaggi Fortini s.r.l.	Ravenna	Italy	Mozzarella di Giovanni	35
Norske Meierier	Sandvika	Norway	Fløtemysost	22
Svensk Sjöföda AB	Stockholm	Sweden	Röd Kaviar	15
Aux joyeux ecclésiastiques	Paris	France	Chartreuse verte	18
Leka Trading	Singapore	Singapore	Gula Malacca	19
Lyngbysild	Lyngby	Denmark	Spegesild	12
Zaanse Snoepfabriek	Zaandam	Netherlands	Chocolade	13
Karkki Oy	Lappeenranta	Finland	Lakkalikööri	18
G'day, Mate	Sydney	Australia	Perth Pasties	33
Ma Maison	Montréal	Canada	Pâté chinois	24
Pasta Buttini s.r.l.	Salerno	Italy	Ravioli Angelo	20
Escargots Nouveaux	Montceau	France	Escargots de Bourgogne	13
Gai pâturage	Annecy	France	Camembert Pierrot	34
Forêts d'érables	Ste-Hyacinthe	Canada	Tarte au sucre	49

10. Kemudian tambahkan tSortRow seperti berikut:

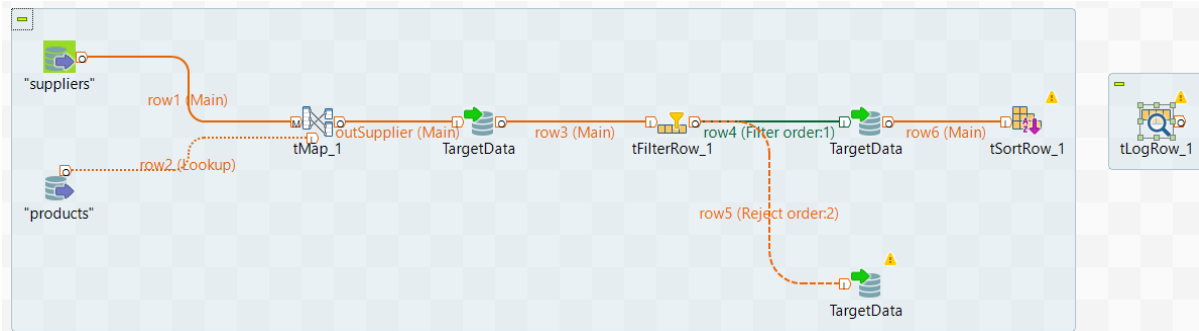


Lakukan pengaturan sortir Kota/City dari A-Z pada tools tSortRow seperti berikut:

Built-In Edit schema Sync columns

Schema column	sort num or alpha?	Order asc or desc?
City	alpha	asc

11. Tambahkan tLogRow dan berikan relasi seperti berikut:



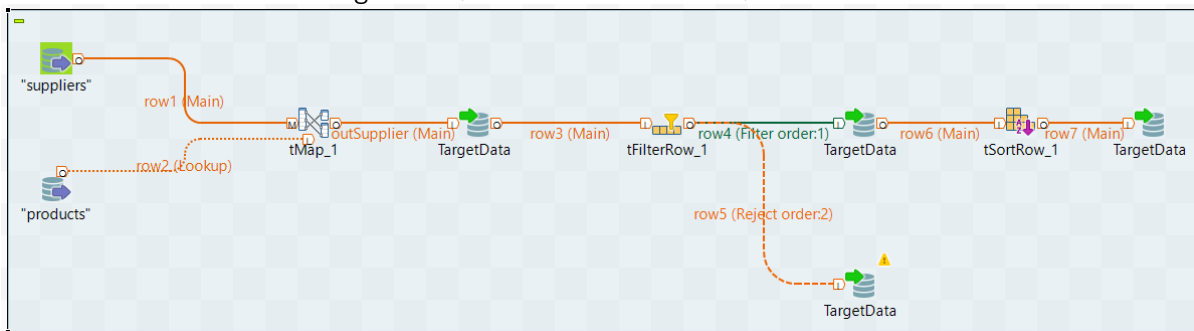
Lakukan Run dan Pastikan hasilnya seperti berikut:

Execution

Run Kill Clear

tLogRow_1				
SupplierName	City	Country	ProductName	Price
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40
Bigfoot Breweries	Bend	USA	Laughing Lumberjack Lager	14
New England Seafood Cannery	Boston	USA	Jack's New England Clam Chowder	10

12. Kemudian Tambahkan TargetData (nama tabel disesuaikan).



Lakukan Run dan Pastikan hasilnya seperti berikut:

Extra options

SupplierName	City	Country	ProductName	Price
Grandma Kelly's Homestead	Ann Arbor	USA	Northwoods Cranberry Sauce	40
Bigfoot Breweries	Bend	USA	Laughing Lumberjack Lager	14
New England Seafood Cannery	Boston	USA	Jack's New England Clam Chowder	10
Exotic Liquid	Londona	UK	Aniseed Syrup	10
Specialty Biscuits, Ltd.	Manchester	UK	Scottish Longbreads	13
New Orleans Cajun Delights	New Orleans	USA	Louisiana Hot Spiced Okra	17

Buatlah Laporan Praktikum (semua Latihan dan Tugas) serta kendala saat mengerjakan praktikum. Kumpulkan pada pada halaman pengumpulan Laporan Praktikum dengan ekstensi .docx